

```

rajesh@rajesh-TravelMate-4150:~$
2 packets transmitted, 0 received, 100% packet loss, time 1007ms

rajesh@rajesh-TravelMate-4150:~$ ping 192.168.1.10
PING 192.168.1.10 (192.168.1.10) 56(84) bytes of data:
64 bytes from 192.168.1.10: icmp_seq=1 ttl=128 time=102 ms
^C
-- 192.168.1.10 ping statistics --
1 packets transmitted, 0 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 102.731/102.731/102.731/0.000 ms
rajesh@rajesh-TravelMate-4150:~$ sudo nmap -n -SS 192.168.1.10
[sudo] password for rajesh:
Starting Nmap 6.08 ( http://nmap.org ) at 2013-03-31 20:29 IST
Nmap scan report for 192.168.1.10
Host is (0.0034s latency).
Not shown: 997 filtered ports
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
MAC Address: 00:08:07:99:C0:92 (Intel)

Nmap done: 1 IP address (1 host up) scanned in 4.73 seconds
rajesh@rajesh-TravelMate-4150:~$

```

Figure 3: nmap scan result for winxp.conf honeypot

```

rajesh@rajesh-TravelMate-4150:~$ sudo nmap -pi-61133 -n 192.168.1.11
Starting Nmap 6.08 ( http://nmap.org ) at 2013-03-31 20:37 IST
Nmap scan report for 192.168.1.11
Host is up (0.0034s latency).
Not shown: 9533 filtered ports
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
80/tcp    open  http
1012/tcp  open  asa-ssl-prtn
MAC Address: 08:11:22:11:55:00 (Climax)

Nmap done: 1 IP address (1 host up) scanned in 164.42 seconds
rajesh@rajesh-TravelMate-4150:~$

```

Figure 4: Scada honeypot nmap scan

how honeyd can be configured to mimic Windows XP SP1.

Configuring honeyd to mimic Windows XP SP1

Create the configuration for the Windows XP honeypot in the winxp.conf file as follows:

```

set winxp personality "Microsoft Windows XP Professional SP1"
set winxp default tcp action block
set winxp default udp action block
set winxp default icmp action reset
set winxp uptime 1234567
add winxp tcp port 135 open
add winxp tcp port 139 open
add winxp tcp port 445 open
set winxp ethernet "intel"
dhcp winxp on eth0
# To configure static IP on eth0, comment the dhcp command and
enable bind command as follows:
# bind ipaddress winxp

```

Explanation of important configuration options: Remember that the first three bytes of the MAC address denote the manufacturer's ID number. The command:

```
set winxp ethernet "intel"
```

...configures MAC address belonging to Intel

Semiconductor to the honeypot. The other commands are self-explanatory. To start the honeypot configured in winxp.conf under daemon mode, use the following command:

```
sudo honeyd -d -f winxp.conf
```

Using the daemon mode will enable you to see all the network requests and corresponding responses on the screen of the honeypot system.

Figures 3 and 4 show the results of scanning Windows' honeypot by nmap.

The important points to consider in the screenshots in these figures are:

- The process is being demoted from root level
- IP address is received from the DHCP server
- The establishment of ARP binding to Intel's MAC address
- Various ARP responses
- UDP port 17500, which is used by Dropbox for local file synchronisation. There is at least one system using Dropbox in this network. The honeypot responds to it by blocking the connection, since UDP port requests are blocked.

The SCADA honeypot

As mentioned on the website <http://scadahoneynet.sourceforge.net/>, the SCADA honeynet project was launched

with the aim of determining the feasibility of building a software-based framework to simulate a variety of industrial networks such as SCADA, DCS, and PLC architectures. It can be used to:

- Build a honeynet for attackers, in order to gather data on attacker trends and tools
- Provide a scriptable industrial protocol simulator to test a real, live protocol implementation
- Research countermeasures, such as device hardening, stack obfuscation, reducing application information, and the effectiveness of network access controls

The project dates way back to 2005 but it is very relevant even today in the challenging SCADA security scenario.

SCADA honeypot installation

Download the latest release of the SCADA honeynet project and expand the tgz to get four Python scripts. The names indicate services emulated by the corresponding scripts:

```

honeyd-ftp.py
honeyd-html.py
honeyd-modbus.py
honeyd-telnet.py

```

Place these files in the `/usr/share/honeyd/plc` folder. Also make sure you have installed Python on your Ubuntu box.

Configuration

Create configuration for the SCADA honeypot in `scada.conf`